SSLC Model Examination, February - 2026:

Chemistry Answer Key

1	Answer: (c)	1
2	Answer: (c)	1
3	Answer: (d) (r, s, q)	1
4	Answer: (c)	1
5	Option A:	
	(a) 7.	1
	(b) 3.	1
	Option B	
	(a) 4 carbon atoms.	1
	(b) But-1-yne: $CH\equiv C-CH_2-CH_3$ or But-2-yne: $CH_3-C\equiv C-CH_3$.	1
6	(a) Group 1	1
	(b) 11	1
7	(a) $ZnCO_3$	1
	(b) Calcination	1
8	(a) Ammonium Chloride (NH_4Cl) and Calcium Hydroxide ($Ca(OH)_2$)	1
	(b) Liquor Ammonia	1
9	Option A	
	(a) Chlorine gas (Cl_2).	1
	(b) $Na^+ + 1e^- \rightarrow Na$.	1
	Option B	
	(a) Iron bangle	1
	(b) $Cu \rightarrow Cu^{2+} + 2e^-$	1
10	(a) 112 L.	1
	(b) Avogadro's Law (Equal volumes of all gases under the same conditions of temperature and pressure contain equal number of molecules).	1
11	(a) Pineapple	1
	(b) Butanoic acid ($CH_3-CH_2-CH_2-COOH$).	1
12	(a) Rate of forward reaction equals rate of backward reaction. At Equilibrium both reactants and products co exist	1
	(b) (i) Adding O_2: Increases forward rate.	1
	(ii) Increasing Pressure: Increases forward rate	1

13 Option A

(a) Sodium Aluminate ($NaAlO_2$). 1

(b) By adding freshly prepared $Al(OH)_3$ and diluting the solution. 1

(c) Calcination or Heating strongly 1

Option B

(a) Coke (C) and Limestone ($CaCO_3$). 1

(b) Carbon Monoxide (CO). 1

(c) $Fe_2O_3+3CO \rightarrow 2Fe+3CO_2$. 1

14 (a) Mg (Magnesium). 1

(b) From Magnesium to Zinc 1

(c) $Mg+Zn^{2+} \rightarrow Mg^{2+}+Zn$ 1

15 (a) Vinyl Chloride $CH_2=CHCl$. 1

(b) Polyvinyl Chloride (PVC). 1

(c) $nCH_2=CHCl \rightarrow -[CH_2-CHCl]_n-$ 1

16 Option A

(a) 2 moles. 1

(b) 10 mol O_2 1

(c) 224 L 1

Option B

(a) 400 g. 2

(b) 176 g. 1

17 (a) MnO_2. 1

(b) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3$ (or [Ar] $3d^3$). 1

(c) Small energy difference between penultimate d and outermost s subshells allowing electrons from both to participate in bonding. 1

18 Option A

(a) $CH_3-CH_2-O-CH_2-CH_3$ 1

(b) Alkoxy group (Ether). 1

(c) Methoxypropane (CH3−O−CH2−CH2−CH3). 1

(d) Butan-1-ol (CH3−CH2−CH2−CH2−OH) or any Butanol isomer. 1

Option B

(a) Pentanal: $CH_3-CH_2-CH_2-CH_2-CHO$. 1

(b) Aldehydes. 1

(c) Pentan-2-one: $CH_3-CO-CH_2-CH_2-CH_3$ 1

(d) Pentan-3-one: $CH_3-CH_2-CO-CH_2-CH_3$ 1

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